Human Pose Detection/Keypoint Estimation using MMVR Dataset

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@inproceedings{rahman2024mmvr,
title={MMVR: Millimeter-wave multi-view radar dataset and benchmark for indoor perception},
author={Rahman, M Mahbubur and Yataka, Ryoma and Kato, Sorachi and Wang, Pu and Li, Peizhao and Cardace, Adriano and Boufounos, Petros},
booktitle={European Conference on Computer Vision},
organization={Springer}
}

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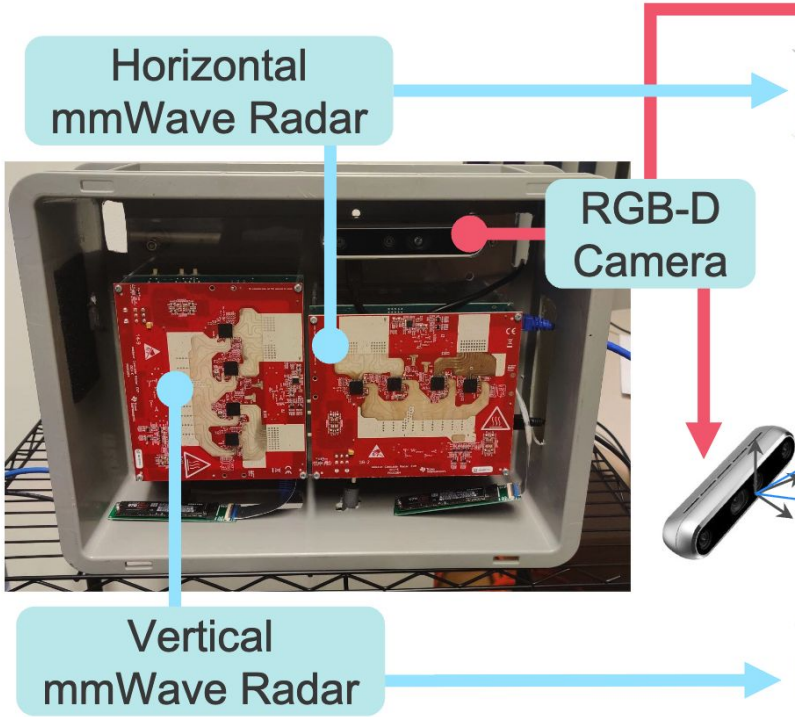
Outline

- Motivation & Problem Statement
- Dataset: MMVR
- Model Architecture: Custom CNN
- Results & Metrics
- Model Comparison: ResNet-18 vs Custom CNN
- Challenges

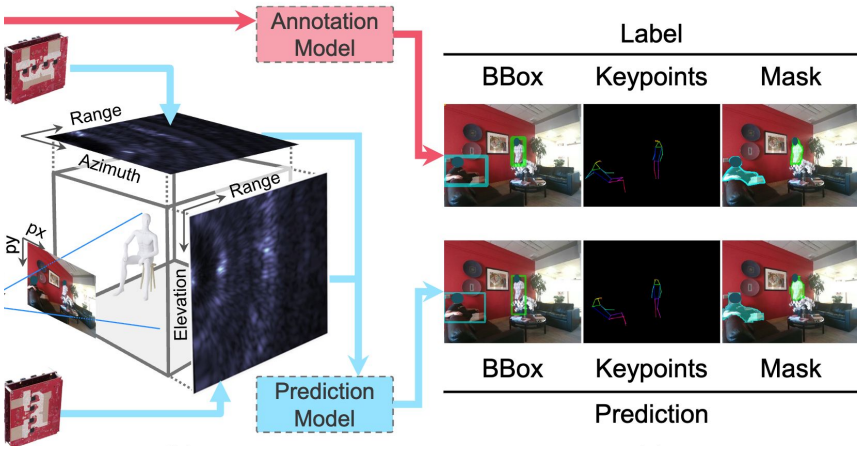


Motivation & Problem Statement

- mmWave radar operates independently of lighting conditions
- Radar adds depth and motion cues beyond 2D imagery
- Privacy preserving

Dataset: MMVR

- https://zenodo.org/records/12611978
- 345,000 multi-view radar heatmap frames
- 25 subjects across 6 rooms over 9 days
- Supports detection, pose estimation, and segmentation tasks
- **Extract downloaded Zip for data-loading module**
- **Do not download the whole dataset.**



Dataset: MMVR

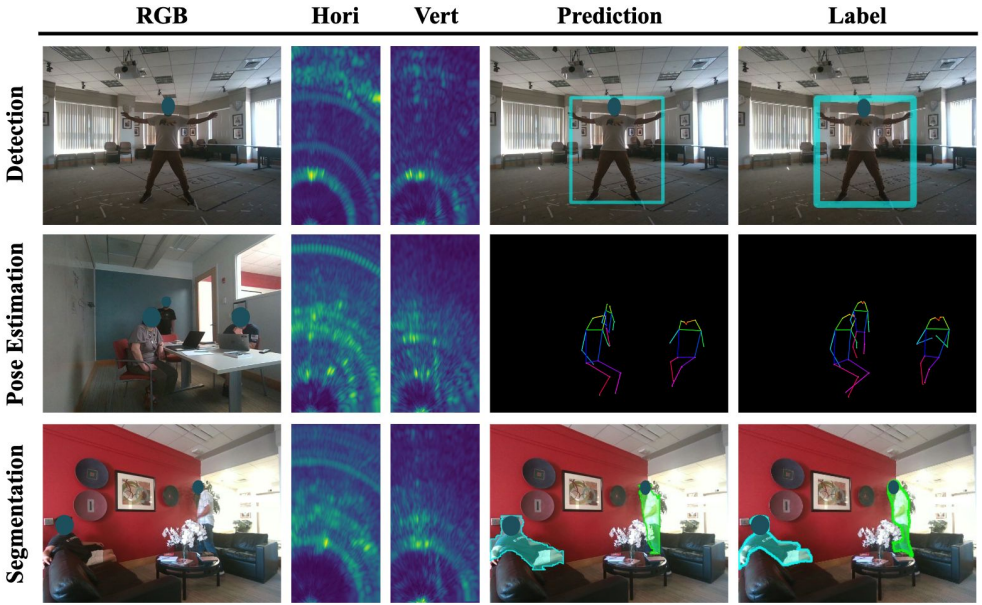


Figure 7: Visualization of data collection (**P1: open foreground** in the first row and **P2: cluttered space** in the second and third rows), RGB (first column), corresponding multi-view (horizontal and vertical) radar heatmaps (second and third columns), baseline prediction results (fourth columns), and annotation labels (fifth columns).

Model Architecture (Hint)

- **Develop CNN-based encoder-decoder with residual and attention blocks**
- Decoder: 2 convolutional layers reconstructing spatial detail
- 1×1 convolution projects features to 17 keypoint heatmaps
- Soft-Argmax layer provides differentiable coordinate prediction
- Output: 17 keypoint (x, y) coordinates per frame
- **Compare your result with ResNet-18**

Results Metrics

- Accuracy
- Mean Absolute Error (pixel distance)
- F1-Score
- Precision
- **Detection metrics:** Percentage of Correct Keypoints (PCK) with 5% of image size
- Compare with baseline model

Challenges

- Radar-to-heatmap transformation is computationally expensive
- Training might require GPU
- Multiple runs will need to determine optimal parameter

Questions

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